BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Network Security Scenarios

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What is penetration testing?

2. What is TCP's role?
   A) Choosing display fonts
   B) Reliable delivery in order
   C) Drawing website images
   D) Assigning MAC addresses

3. What is HTTP used for?

4. A pupil logs in to a website. "HTTPS proves every website is honest." Correct this. What does HTTPS protect?

Today's objectives

Recognise relevant networking/security issues in a scenario. Match a threat to an appropriate control.

Key vocabulary - Read, cover, recall

Confidentiality
Keeping information private from unauthorised readers.

Integrity
Keeping data accurate and protected against unauthorised change.

Availability
Making systems and data accessible to legitimate users.

Phishing
Deceptive message seeking sensitive information.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

Start with the symptom

Read the scenario and identify what is being targeted: confidentiality, integrity or availability.

Name the threat

Use exact J277 terminology: phishing, malware, brute-force, interception, DoS or SQL injection.

Match the defence

Choose a measure that counters the threat: education, updates, anti-malware, access rights, encryption, authentication, lockout or firewall.

Explain the mechanism

An exam-quality answer says what a control does and why that interrupts the particular attack. Avoid simply listing security products.

I Do - Worked model

A pupil receives a fraudulent email asking them to sign into a fake school portal. Explain the threat and TWO ways to reduce the risk. [4 marks]

- The email pretends to be trusted and asks for login credentials.
- Phishing is a deceptive message intended to steal account details [1].
- Training users to spot false links and unusual requests [1] can reduce unsafe clicks [1].
- Two-factor authentication adds another check [1] so a stolen password alone may not allow access. Four teacher marking points are distributed across the explanation.

We Do - Guided practice

1. A school account is receiving repeated incorrect passwords. Explain two security controls that limit the attack. [4 marks]

2. A laptop downloads harmful software. Explain two defences that could reduce damage. [4 marks]

3. A school's online service is overloaded with requests. Identify the threat and explain one suitable defence. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Change alertLevel to 1. What message prints?

```
alertLevel = 2
if alertLevel > 1 then
print("Tell IT")
else
print("Continue checks")
endif
```

You Do - Independent exam questions

Q2. A message impersonates a bank and requests a password. Name the threat. [1 mark]

Q3. An employee uses unprotected public Wi-Fi for confidential information. Explain the risk and a suitable control. [4 marks]

Q4. A school faces phishing, malware and brute-force login attempts. Describe one suitable protection for each and how it helps. [6 marks]

Exit Ticket - True or False?

Choosing security measures should depend on the particular threat described.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Testing with permission, within agreed limits, to find and fix security weaknesses.

2. B - Reliable delivery in order

3. Transferring web resources between web clients and servers.

4. HTTPS encrypts data in transit. It does not prove the website or its owner is honest.

I Do - Explain the model

The email pretends to be trusted and asks for login credentials.

Phishing is a deceptive message intended to steal account details [1].

Training users to spot false links and unusual requests [1] can reduce unsafe clicks [1].

Two-factor authentication adds another check [1] so a stolen password alone may not allow access. Four teacher marking points are distributed across the explanation.

We Do - Model answers

1. An account lockout stops repeated attempts after a threshold [1], limiting brute-force guesses [1]. Two-factor authentication requires another proof of identity [1], so a guessed password alone is insufficient [1].

2. Anti-malware detects/quarantines the threat [1], reducing malicious activity [1]. Restricted user access rights prevent ordinary accounts making system-wide changes [1], limiting potential damage [1].

3. It is a denial-of-service attack [1] using excessive traffic/requests [1]. Firewall/rate limiting can filter or restrict unwanted traffic [1], helping keep service available to legitimate users [1].

Code First - Answer

```
alertLevel = 1
if alertLevel > 1 then
print("Tell IT")
else
print("Continue checks")
endif
```

Expected output: Continue checks

You Do - Self-mark

Q2: Phishing/social engineering [1].

Q3: Data could be intercepted during transmission [1] and private details read [1]. Encryption protects content in transit [1] so an interceptor without the key cannot understand it [1].

Q4: Phishing: awareness training [1] reduces risky link clicks [1]. Malware: anti-malware software [1] detects/quarantines malicious programs [1]. Brute-force: account lockout [1] limits repeated password attempts [1]. Other correctly justified measures acceptable.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.